

WS5 — A census of the one-schema VAL(2) family

The first machinery in this program that produces machines rather than consuming them

1. Why this family

WS4 closed by asking whether the one-schema valuation class is Turing-complete, and observing that it is the only two-sided bet the program still holds. That class is also where both flagship machines live. So a census of it does two jobs at once: it populates the class whose power is open, and it converts per-machine handcraft into a pipeline.

A machine is five integers. Write $x = 2^v m$ with m odd. If $m = 1$ the machine **halts** (x is a pure power of two). Otherwise $m = 2k+1$, so $x = 2^{v+1}k + 2^v$, and

$$F(x) = A_v k + B_v, \quad A_v = \alpha \cdot 2^{v+1} + \beta, \quad B_v = \gamma \cdot 2^v + \delta \cdot v + \varepsilon$$

— the branch-affine normal form the whole program is built on, with the coefficients left free. **The Space Needle is (1, 3, 1, 1, 0)**, verified: zero mismatches against its own step function over $2 \leq x < 300,000$, and the orbit from $x_0 = 3$ runs 3, 6, 10, 17, 41, 101, 251, ..., joining the published orbit at 6. As a function of x the branch has slope $A_v \sqrt{2^{v+1}} \rightarrow \alpha$, so α is the asymptotic multiplier: $\alpha \geq 2$ doubles at every branch, $\alpha = 1$ is the weakly expanding Collatz-like regime.

Update, July 31 — the box holds a second wild machine. Member **(1, 1, 1, 1, 0)** turns out to be the generic branch of the **sheep machine** (1RB1LA_0LC0RC_1LE1RD_1RE1RC_1LF0LA_ - - -1LE, BB(6), found by *sheep* on 7 April 2026 and listed as a bbchallenge Cryptid): the Space Needle's reduction with β changed from 3 to 1. Two members of this box are therefore reductions of machines other people found, and the family is less manufactured than it looked. There is a sting in it. The census records (1,1,1,1,0) as **HALT** — from $x = 3$ it steps $3 \rightarrow 4 = 2^2$. The sheep machine has one *exceptional* branch, at oddPart = 3, which this schema cannot express, and it intercepts exactly that value: $3 \rightarrow 6$. **One extra branch on one odd part converts a halting census member into an open problem.** That is the clearest available statement of what the box misses, and it is an argument for widening the schema rather than the parameter ranges.

2. What every machine gets, and the one part that is new

Well-definedness, simulation from $x_0 = 3$, drift, the backward branching ceiling, the WS3 forbidden-branch sieve, and an exact congruence decision. Two of those can decide a machine outright, which is the point.

The congruence test here is exact and complete, not the necessary condition WS4 had to settle for. On branch v the pair (source, target) mod m traces the graph of a single affine map ϕ_v , and A_v , B_v mod m depend on v only through the pair $(2^v \bmod m, v \bmod m)$ — a state space of size m^2 . So iterating v until that state repeats enumerates every branch, the relation R_m is exactly computable, the orbit lies inside the R_m -closure of x_0 , and if that closure misses the halting set's residues the machine **provably** never halts.

Calibration ran first, in both directions, because a decision procedure reported without both is worthless: the Space Needle must *not* be decided (it is not), and machines that *are* decided must survive an independent check (they do).

3. What came out

1,080 machines enumerated over $\alpha \in \{1,2,3\}$, $\beta \in \{-1..7\}$, $\gamma \in \{1,2,3\}$, $\delta \in \{0,1,2\}$, $\varepsilon \in \{-2..2\}$; **3 are not well defined** (F leaves the positive integers); **1,077 analysed** in 196.3 seconds.

α	machines	HALT	CYCLE	GROW	decided	mean drift
1	357	116	21	220	105	0.8809
2	360	80	6	274	103	1.5529
3	360	61	0	299	110	1.9933

318 of 1,077 machines — 29.5% — are decided by a separating congruence, i.e. proved never to halt from $x_0 = 3$. The deciding moduli run 3 (41), 4 (105), 6 (68), 7 (3), 8 (18), 10 (34), 12 (17), with a tail out to 56.

Every one of the 318 certificates was re-audited independently — brute force over real integers to $x < 200,000$, powers of two enumerated completely, without reusing the branch enumeration that produced the certificate — with **0 failures**. A false proof of non-halting is the worst output this program could emit, so this audit is not optional.

3.1 G2 finally moves, and here is exactly how much

G2 — decide a cryptid — had not moved in four consecutive stock-takes. It moves here: 318 machines decided, and two proved non-halting *from every start* (Section 4). **The honest qualification is that these are new and easier machines, not the cryptids**; a machine with a separating congruence is by definition not one. What the census delivers is the partition — which members are easy, why, and a pool of hard ones.

It is worth noting what it cost. 196.3 seconds of census decided 318 machines. Two days of SAT search bought two states.

4. The first three theorems

T1 (proved). The machine **(1, 1, 2, 0, 1)** never halts after its first step. Here $B_v = 2 \cdot 2^v + 1 = 2^{v+1} + 1 = A_v$, so $F(x) = A_v(k+1)$ with A_v odd and at least 3. Every image carries an odd factor greater than one and is therefore never a power of two.

T2 (proved). The machine **(2, -1, 2, 1, 1)** never halts after its first step. $A_v = 2^{v+2} - 1$, and the WS3 sieve forbids a halt out of branch v unless $Q_0 2^e \equiv P_0 \pmod{A_v}$ with $P_0 = (2v+3)2^v$ and $Q_0 = 1 - 2^{v+1}$. Modulo A_v we have $2^{v+2} \equiv 1$, hence $2Q_0 \equiv 1$, so Q_0 is the inverse of 2 and the condition collapses to

$$2^{e+1} \equiv 2v + 3 \pmod{2^{v+2} - 1}$$

The powers of two mod A_v are exactly $\{1, 2, \dots, 2^{v+1}\}$, since 2 has order $v+2$; and for $v \geq 1$ the number $2v+3$ is odd, greater than 1 and less than A_v , so it is its own residue and is not among them.

For $v = 0$, $A_0 = 3$ and $2v+3 \equiv 0$, also not a power of two. **Every branch is forbidden**, so no orbit can reach a power of two after a step. **QED**

T3 (proved, and exact on the whole census). A member is decided at $m = 3$ — hence provably never halts from $x_0 = 3$ — if and only if

$$\delta \equiv 0, \quad \gamma \equiv \alpha, \quad \beta + \varepsilon \equiv 0 \quad (\text{all mod } 3)$$

Proof. Mod 3 the powers of two are $\{1, 2\}$, every nonzero class, so a separating class must be contained in $\{0\}$; it contains $x_0 = 3$, so it is $\{0\}$. Then $3 \mid x$ forces $k \equiv 1 \pmod{3}$, and $F(x) \equiv (-1)^V(\gamma - \alpha) + \beta + \delta v + \varepsilon$. Vanishing at $v = 0, 1, 2$ gives $2\delta \equiv 0$, then $2(\gamma - \alpha) \equiv 0$, then $\beta + \varepsilon \equiv 0$; conversely those three make it vanish for every v . **QED** Checked against the census: 41 predicted and decided, 0 predicted but not decided, 0 decided but not predicted, over 1,077 machines.

Corollary, and the sharpest thing the census says about the Needle. The **v-linear term δ is what blocks the cheapest certificate there is**. The Space Needle has $\delta = 1$ and fails T3 in one line. The term δv injects unbounded valuation information into the *value*, and no modulus can track it.

5. The Needle is not special in anything measured globally

Drift and the branching ceiling are functions of (α, β) alone — verified: 24 (α, β) classes, and **none** carrying more than one (drift, ceiling) value. So **45 machines share the Needle's drift 0.9411 and ceiling 0.54528 exactly**. Neither statistic can tell it apart from 44 siblings.

What does separate them is the sieve, which reads $(\gamma, \delta, \varepsilon)$: across those twins the forbidden-branch mass ranges 0.0000 to 0.7988, with the Needle at **0.2852** — a clean reproduction of WS3's independently measured 28.7% (that was over $v \leq 35$, this over $v \leq 8$).

β parity is what makes a machine hard. Even β : 282/405 decided (70%). Odd β : 36/672 (5%). Of the 120 cryptid candidates ($\alpha = 1$, grows, undecided), **108 have odd β** . A_v odd is exactly the classic Collatz situation: the map hands you no 2-adic structure to exploit.

6. Every branch forbidden: seven machines, then ten

Seven undecided growers have *all* branches sieved out to $v = 200$ (mass 1.000000000). Two of them became T1 and T2 on the day. **The other five had no uniform argument and were not claimed** — “forbidden to $v = 200$ ” is not “forbidden for all v ”, and that gap is exactly the kind this program refuses to paper over. All five are now theorems, along with three machines this search had never seen; and the count “seven” turns out to be an artefact of the pool the leads were drawn from.

6.1 The sieve lemma is universal

The first follow-up derived $S_v \equiv 2B_v$ “for $\alpha = 2, \beta = -1$ ”, and closed three leads with it. **That attribution was too modest: the derivation never uses either value.** On branch v the sieve forbids a halt unless $Q_0 \cdot 2^e \equiv P_0 \pmod{A_v}$, where $P_0 = (2B_v - A_v)2^v$ and $Q_0 = 2^{v+1} - A_v$ come from the branch's affine fixed point. Reduce both mod A_v and the A_v terms simply vanish: $Q_0 \equiv 2^{v+1}$ and $P_0 \equiv B_v \cdot 2^{v+1}$, so $P_0 Q_0^{-1} \equiv B_v$.

Universal sieve lemma. For every machine with β odd: $S_v \equiv 2B_v \pmod{A_v}$

Equivalently, and this is the form worth remembering: **branch v can immediately precede a halt if and only if B_v lies in $\langle 2 \rangle \pmod{A_v}$** , the multiplicative group generated by 2. *Machine-verified: 672 machines — the whole odd- β box — 51030 branches, $v = 0$ to 80, no mismatches.*

And because $A_v = \alpha 2^{v+1} + \beta$ gives $\alpha 2^{v+1} \equiv -\beta$, multiplying through by α kills the exponential term outright:

Linear corollary. $\alpha S_v \equiv 2\alpha\delta v + 2\alpha\varepsilon - \beta\gamma \pmod{A_v}$

— affine in v , for every machine in the family (verified on the same branches, no mismatches). **The exponential is not in the target; it is only in the modulus.** That single sentence is the whole shape of the difficulty: what has to be decided is whether a linear function of v lands in the group generated by 2 modulo a number that doubles with v .

So what decides a machine is the **size of $\langle 2 \rangle$** . Four classes are **listable** — $\text{ord}(2)$ grows linearly, so the powers of two are a short explicit list a parity/size argument can clear: $(2, -1)$ with $\text{ord} = v + 2$ and $\langle 2 \rangle = \{2^i\}$; $(1, 1)$ and $(2, 1)$ with $\text{ord} = 2(v+1)$ and $2(v+2)$, giving the signed list $\{\pm 2^i\}$; and $(3, 3)$, where 3 divides A_v . The rest are **thin**: ord grows exponentially on average — $(1, 3)$ runs 4, 3, 10, 18, 12, 66, 130, 36 — and membership has no elementary handle at all.

6.2 The ten machine theorems (T3 is the $m = 3$ criterion)

Every all-branches-forbidden machine in the box, with S_v as an **exact integer identity** (asserted to $v = 4,000$ on a second, independent code path) and the pattern that finishes it. T7 to T11 are new; the two leads previously called “the ones that do not transfer” are T7 and T8.

machine	S_v (derived)	why every branch is forbidden	
(1,1,2,0,1)	0	$B_v = A_v$ exactly; 0 is not a unit and every power of two is	T1
(2,-1,2,0,1)	3	$2B_v = A_v + 3$; odd, $1 < 3 < A_v$ for $v \geq 1$	T9
(2,-1,2,1,1)	$2v + 3$	odd, $1 < S_v < A_v$ for $v \geq 1$; $v = 0$ gives 0	T2
(2,-1,2,2,1)	$4v + 3$	odd, in range for $v \geq 2$; $v = 0, 1$ give 0	T4
(2,-1,3,0,0)	$2^{v+1} + 1$	odd, in range for $v \geq 1$; $v = 0$ gives 0	T5
(2,-1,3,1,0)	$2^{v+1} + 2v + 1$	odd, in range for $v \geq 2$; $v = 0, 1$ give 0	T6
(2,-1,1,2,-1)	$2^{v+1} + 4v - 2$	$= 2 \cdot \text{odd}$ and $< A_v$, so $S_v = 2^i$ forces $i = 1$, i.e. $2^v = 2 - 2v$	T10
(2,1,1,2,-1)	$2^{v+1} + 4v - 2$	same value, signed list mod $2^{v+2}+1$; $S_v = A_v - 2^i$ forces $i = 0$, i.e. $4v - 2 = 2^{v+1}$	T7
(3,3,2,0,1)	$\equiv 0 \pmod{d_v}$	$B_v = d_v = 2^{v+1}+1$ exactly and d_v divides A_v ; powers of two are units mod d_v	T8
(3,3,3,0,0)	-3	$2B_v = A_v - 3$ and 3 divides A_v , so $S_v \equiv 0 \pmod{3}$; powers of two mod 3 are $\{1, 2\}$	T11

T7 to T11 (proved): five more machines that never halt after their first step. Brute force on all ten — every non-power-of-two start below 20,000, 400 steps — finds 0 halts.

Three of the five were invisible to the earlier hunt **not because the test differed but because of the pool filter**. The leads were drawn from undecided *growers*: (2,-1,2,0,1) and (3,3,3,0,0) were already congruence-decided, and (2,-1,1,2,-1) is a CYCLE machine ($x = 3$ is a fixed point, $F(3) = 3$). For the two congruence-decided ones this is a real upgrade — the congruence gave “never halts from $x_0 = 3$ ”, the sieve gives “no orbit from *any* start reaches a power of two after a step”. A filter chosen for one purpose quietly bounded the reach of another.

6.3 The completeness theorem: the method is finished

Those ten are all there are. For every one of the other machines in the odd- β box, the sweep returns an **explicit surviving branch**: a $v \leq 22$ together with the exponent e witnessing $2^e \equiv S_v \pmod{A_v}$. *Machine-verified: 672 machines swept — 10 all-forbidden, 620 carrying a surviving-branch certificate, 42 sieve-silent* (the (1,-1) class, where no branch expands and the sieve argues from nothing).

A surviving branch is a **positive certificate**, not a failure to find one. So this is not exhaustion in the sense of the meta report’s bounded-resource bounded-resource searches — it is a proof that the sieve-to-theorem pipeline is **complete on this family, at ten machines**. It is the first time this program has closed a method with a completeness statement rather than a budget. What is *not* claimed: a surviving branch does not make a machine halt. It means only that this method cannot decide it, and something else must.

6.4 The frontier map, and where the Needle sits

Of the 636 odd- β machines not decided by a congruence:

tier	count	what it means
1 — mechanical	8	all branches forbidden, with a pattern proof: the ten, less those already decided another way
2 — S-unit	399	thin $\langle 2 \rangle$; the sieve bites but does not close. Deciding these needs S-unit or Baker input, not more of this
3a — sieve-closed	164	listable class, but a branch certifiably survives: this route is provably shut
3b — sieve-void	0	thin, forbidden mass 0 — every tested branch survives
3c — sieve-silent	41	the (1,-1) class: no expanding branch anywhere

The Space Needle is in tier 2, and 120 machines share its group. With $A_v = 2^{v+1} + 3$ we get $2^{v+1} \equiv -3$, so $\langle 2 \rangle = \{\pm 3^a \cdot 2^i\}$ and the halting condition on branch v reads:

$$2v - 3 \in \{\pm 3^a \cdot 2^i\} \pmod{2^{v+1} + 3}$$

— a 2,3-S-unit membership question, and that is now the whole of the difficulty, stated in one line. *Verified: the closed form $S_v \equiv 2v - 3$ is exact for $v \leq 200$ with no mismatches; the weighted forbidden mass is 0.2868, an independent reproduction of WS3’s 28.7%; and 19 branches are forbidden among the first 35 valuations, matching WS3’s count exactly.*

And it does not close. The branches that survive to $v = 40$ are 0, 2, 3, 5, 6, 8, 11, 14, 15, 16, 17, 20, 27, 29, 30, 34, 37, 38 — 18 of 41, and **they are not thinning out**. There is no asymptotic

all-branches-forbidden theorem waiting here, and the last-step sieve saturates near 28.7% of weight forever. Whatever decides the Needle comes from somewhere else. Recording that saves the next reader the search.

7. Status of every claim

claim	epistemic status
The family contains the Space Needle at (1,3,1,1,0)	machine-verified (0 mismatches, $x < 300,000$)
318 machines never halt from $x_0 = 3$	proved — the congruence test is exact and complete for its class; all 318 certificates independently audited, 0 failures
T1, T2, T4 to T11: ten machines never halt after their first step	proved — and the wording matters: a start that <i>is</i> a power of two halts at step 0, so the powers of two are exactly the halting starts. Brute-forced: non-power-of-two starts $< 20,000$, 400 steps, 0 halts on all ten
The sieve lemma $S_v \equiv 2B_v$ holds for every β odd, and αS_v is affine in v	proved; machine-verified on 51030 branches of 672 machines, $v \leq 80$, no mismatches
Exactly ten machines in the box have every branch forbidden	proved — every other machine carries an <i>explicit surviving branch</i> , a positive certificate, so this is completeness and not exhaustion (672 swept, 42 sieve-silent)
T3: closed form for the $m = 3$ certificate	proved, and exact on all 1,077 machines
Drift and ceiling depend only on (α, β)	machine-verified over the enumerated range
The Needle's branch condition is $2v - 3 \in \{\pm 3^a 2^1\} \bmod 2^{v+1} + 3$	proved (Section 6.4); and the surviving branches do not thin out — 18 of 41 survive to $v = 40$, so no asymptotic theorem is available by this route
The 120 cryptid candidates are cryptids	not claimed. They are expanding with an undecided orbit; no halting-set characterization or equidistribution model has been built, and they are siblings in one family, not independent discoveries
One-schema VAL(2) is / is not Turing-complete	open — untouched by this work. A census maps a family; it does not bound its power

Code: *family.py*, *decide.py*, *census.py*, *verify.py*, *theorems.py*, *leads.py*, *universal.py*, with logs and *census_rows.tsv* in the same directory. Report regenerated by *make_census_report.py*, which reads every number from those files.